

Butterfly Challenge — Standalone Animated Microsite

Project Concept (v5.0)

Date: 2026-05-14

0. EXECUTIVE SUMMARY

The project in one sentence:

We are building a **standalone, cinematic scroll-driven animated microsite** for the Butterfly Challenge movement — a complement to `butterflychallenge.net`, delivering a 4-second "reveal" experience that engages users on an emotional level.

Why this is needed:

- The main site **informs** — the new microsite **makes people feel**
- This is an industry-standard approach used by Apple iPhone reveal, Ford Mustang launch campaigns
- High viral content potential on social media (ideal for sharing on TikTok / Instagram)
- An emotional entry point for the movement

Key numbers:

Metric	Value
Budget	\$15 (1-month AI tool subscription)
Hosting	Vercel (free)
Time	3-4 days
Start date	2026-05-14
Launch date	2026-05-17
Team	Project Owner + Developer + AI Assistant
Will the main site change?	NO — it will not be touched
Scope	Hero/header section only (standalone cinematic experience)

Core technology:

Next.js + GSAP + Canvas (the **Scroll-Driven Image Sequence Animation** technique used by Apple/Ford/BMW).
Asset generation via AI video, then extracted into 240 frames using FFmpeg.

Risk level:

Low-to-medium — proven technology, AI tools are available, small budget. Main risk: AI video quality may not be acceptable on the first attempt (mitigation: multiple iterations).

1. PROJECT GOAL AND POSITIONING

What this is:

The **animated version** of `butterflychallenge.net` — built as a **completely separate** site. Like the Ford "Rediscover the Classics" campaign, it consists of **only a header + hero** cinematic landing experience. The animation plays as the user scrolls, and at the end they are directed to the `.net` site.

What this is NOT:

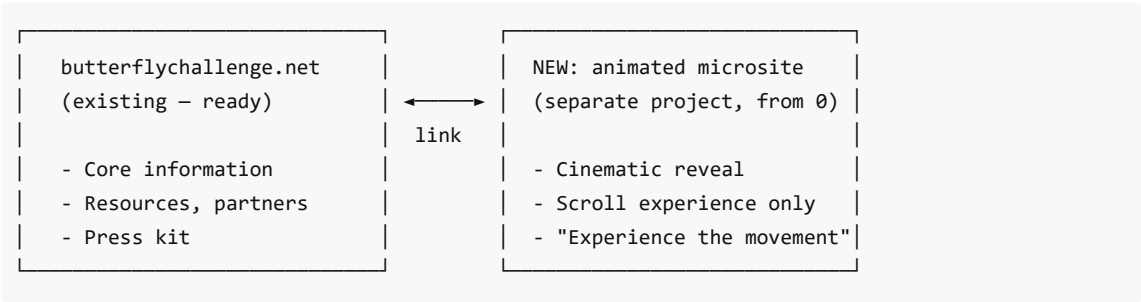
- Not an update to `butterflychallenge.net`
- Not adding animation to the main site
- The main site's code will not be touched
- Text/content from the `.net` site will not be reused

CONTENT POLICY (important):

This animated microsite has **its own dedicated content**. Text from the `.net` site is **not copied**. Only:

- Its own minimal cinematic copy (title + tagline + interaction prompt + CTA)
- Like the Ford reference — the animation itself is the primary "message," text is minimal
- The only connection: the final CTA button links to `butterflychallenge.net`

Relationship:



Primary goal:

In the first 5-10 seconds after a user lands on the page, they should feel the emotional power of the **"Butterfly Sign"** gesture — just like Ford's classic car is mystically revealed from beneath a cover. At the end of the experience — a link to `butterflychallenge.net` .

2. ANIMATION TECHNIQUE (official name)

Scroll-Driven Image Sequence Animation (synonyms: Frame Scrubbing on Scroll • Scrollytelling Pin • Canvas Frame Sequencer)

Mechanism:

1. **240 image frames** are prepared in advance (AI generation)
2. Images are converted from **PNG** → **WebP** (~70% smaller)
3. The current frame is drawn inside an **HTML Canvas**
4. **GSAP ScrollTrigger** maps the scroll position to a 0.00 → 1.00 range
5. This progress value is converted to `frameIndex = Math.floor(progress * 239)`
6. No video element is used — each frame is a separate image

Why not a `<video>` tag with `currentTime` :

- Glitches and stalls on iOS Safari
- No frame-perfect accuracy (keyframe seeking lag)
- Freezes on reverse scroll
- Breaks on fast scrolling
- **Canvas sequence is the industry standard** (Apple, Ford, BMW use this technique)

3. ASSET PRODUCTION PIPELINE

Step-by-step:

Step 1: AI video generation (Runway Gen-3 or similar)

- One continuous 4-second video
- 1080p (minimum) or 4K, 60fps recommended
- Done by the developer

Step 2: Frame extraction (FFmpeg)

```
# Extract frames from the video at 60fps
ffmpeg -i reveal_animation.mp4 -vf fps=60 frames/frame_%04d.png

# Output: frame_0001.png ... frame_0240.png (4 sec x 60fps = 240 frames)
```

Step 3: WebP optimization

```
# Convert each PNG to WebP (70-80% smaller)
for f in frames/*.png; do
  cwebp -q 80 "$f" -o "${f%.png}.webp"
done

# Smaller version for mobile
for f in frames/*.webp; do
  ffmpeg -i "$f" -vf scale=720:-1 "mobile/${basename $f}"
done
```

Step 4: Integration into the site

- Upload to the ScrollSequence Canvas player component
- Display with lazy loading

4. FORD → BUTTERFLY CHALLENGE mapping (1:1)

Original Ford element	Butterfly Challenge equivalent
Classic Mustang car	Human silhouette (in Butterfly Sign pose)
White cloth/sheet (cover)	Silk-like flowing fabric — which transforms into butterflies
Crossed neon line lights	Soft backlit butterfly wing contour (neon glow)

Concrete garage	Open cosmic / infinite horizon background (cosmic dust + bokeh)
"REDISCOVER the CLASSICS"	" THE BUTTERFLY SIGN " (main title, original copy)
Subtitle: "It's a look that has spanned generations"	" A gesture seen. A signal felt. " (short tagline, original)
"DRAG TO REVEAL"	" SCROLL TO REVEAL "
Showing the license plate (APC 420)	Final frame: brand mark / logo
Ford logo	Butterfly mark (image)
"BUILD & PRICE" / "CURRENT OFFERS" buttons	" LEARN MORE " → butterflychallenge.net (the only connection point)

Note: The text strings above are **proposals** — final copy will be written specifically for the animated site; text from the `.net` site will not be used.

5. SCENE-BY-SCENE STORYBOARD (240 frames / 4-second scroll length)

Text note: The strings below are **placeholders** — final copy to be written later. Text from `.net` will **not be used**.

FRAMES 0-40 (Opening — "Wide shot")

- **Camera:** wide, the full scene is visible
- **Subject:** a human silhouette fully covered by fabric, centered
- **Lighting:** soft spotlight from above
- **Background:** dark, starry/misty
- **Text overlay (sticky left):** THE BUTTERFLY SIGN (large serif)
- **Right side:** SCROLL TO REVEAL + circular indicator
- **Bottom (short tagline):** A gesture seen. A signal felt.

FRAMES 40-100 (Zoom-in — closer)

- **Camera:** slowly pushes in toward the silhouette
- **Fabric:** begins to sway in a gentle wind
- **Lighting:** stronger rim light starts to form
- **Text:** the left title fades out (opacity: 1 → 0)

FRAMES 100-160 (Reveal — fabric lifting)

- **Camera:** orbits to the side
- **Fabric:** flies into the air, **begins transforming into butterflies** — each fold becomes a butterfly
- **Text:** no text (the animation's strongest moment — content is not needed)

FRAMES 160-220 (Climax — "Butterfly Sign" revealed)

- **Camera:** focused directly on the hands
- **The person's hands** are visible in the Butterfly Sign pose (thumbs crossed, fingers spread like wings)
- **Around them:** dozens of butterflies flying away (the remnants of the fabric)
- **Lighting:** sharp contrast, hands in golden/warm tones
- **Text (center, fade-in):** keyword (placeholder, e.g., "THIS IS THE SIGN")

FRAMES 220-240 (Calm ending — CTA)

- **Camera:** stops, hands centered
- **Butterfly mark logo** appears at the bottom
- **CTA button enters with animation:**
 - `LEARN MORE` → `butterflychallenge.net` (the only external link)

6. ANIMATION LAYERS (Layered Approach)

Five layers running simultaneously on the page:

LAYER 5: Floating UI elements (nav, logo)	← always visible
LAYER 4: Animated text overlays	← GSAP timeline
LAYER 3: Particle effects (butterflies, dust)	← real-time canvas
LAYER 2: Image sequence (main render)	← scroll-controlled
LAYER 1: Background gradient + bokeh	← CSS animation

Each layer is independent and adds additional life to the main image sequence.

7. TECHNICAL ARCHITECTURE

Project structure:

```
NetAnimation/
├─ /app/
│  ├─ page.tsx                ← Main reveal page
│  ├─ layout.tsx
│  └─ globals.css
├─ /components/
│  ├─ ScrollSequence.tsx      ← Canvas frame player
│  ├─ ButterflyParticles.tsx  ← Particle layer
│  ├─ HeroTextOverlay.tsx     ← Sticky text animation
│  ├─ ProgressIndicator.tsx   ← "SCROLL TO REVEAL" indicator
│  └─ CTABlock.tsx           ← Final CTA
├─ /public/
│  └─ sequence/
│     ├─ desktop/frame_0001.webp ... frame_0240.webp
│     └─ mobile/frame_0001.webp ... frame_0240.webp
├─ /hooks/
│  ├─ useScrollProgress.ts
│  └─ useImagePreloader.ts
├─ /lib/
│  └─ canvas-utils.ts
└─ /scripts/
```

```
| └─ extract-frames.sh          ← FFmpeg video → frames
|   └─ optimize-webp.sh        ← PNG → WebP
└─ package.json
```

Technology stack:

```
Framework:    Next.js 15 (App Router)
Language:     TypeScript
Styling:      Tailwind CSS
Animation:    GSAP + ScrollTrigger
Smooth:       Lenis (smooth scroll)
Canvas:       Native HTML5 Canvas 2D
Particles:    tsParticles or custom canvas
Deploy:       Vercel
```

Core component (pseudocode):

```
function ScrollSequence() {
  const canvasRef = useRef<HTMLCanvasElement>(null);
  const images = useImagePreloader('/sequence/desktop/', 240);

  useGSAP(() => {
    ScrollTrigger.create({
      trigger: '.hero',
      start: 'top top',
      end: '+=400%',
      pin: true,
      scrub: 0.5,
      onUpdate: (self) => {
        const frame = Math.floor(self.progress * 239);
        drawFrame(canvasRef.current, images[frame]);
      }
    });
  });

  return <canvas ref={canvasRef} className="hero-canvas" />;
}
```

8. PERFORMANCE BUDGETS

Metric	Target	Strategy
LCP (Largest Contentful Paint)	< 2.5s	First frame inlined as base64, rest lazy
TTI (Time to Interactive)	< 3.5s	Sequence images deferred load
Total Page Weight	< 15 MB (desktop), < 6 MB (mobile)	WebP + responsive sequences

FPS during scroll	60 FPS	RequestAnimationFrame + offscreen canvas
Memory	< 200 MB	Image bitmap recycling, ImageBitmap API

Fallback strategy:

- **Slow 3G:** Sequence is skipped, a static hero image is shown
- **prefers-reduced-motion:** Animation is disabled, a few static images shown
- **No JS:** SSR fallback with a static hero

9. IMPLEMENTATION PHASES (3-4 days)

The entire project is delivered within **3-4 days**. The site consists of **only a hero/header** — small scope, fast delivery.
All technical work is performed by the developer.

Day 1: Asset generation and setup

Developer.

- ☐ AI video generation (Runway / Sora / Kling) — images and video
- ☐ 4-second high-quality video (1080p, 60fps)
- ☐ Next.js project setup
- ☐ Convert video → 240 PNG frames via FFmpeg (AI assistant script)
- ☐ WebP optimization (AI assistant script)
- ☐ Place assets in `/public/sequence/`

Day 2: Core animation

Developer.

- ☐ ScrollSequence Canvas component
- ☐ GSAP ScrollTrigger integration
- ☐ Image preloader hook
- ☐ Sticky pin + scrub logic
- ☐ Text overlay animations (title, tagline, CTA)

Day 3: Polish, Responsive, Deploy

Developer.

- ☐ Particle layer (butterflies — optional, if time allows)
- ☐ Smooth scroll (Lenis)
- ☐ Mobile sequence (smaller version)
- ☐ Progress indicator
- ☐ CTA block — link to `butterflychallenge.net`
- ☐ Cross-browser test (Chrome, Safari, Firefox, Edge)
- ☐ Mobile test (iOS + Android)
- ☐ Deploy to Vercel

Day 4 (buffer):

- ☐ Final QA and polish
- ☐ Lighthouse audit
- ☐ Analytics integration
- ☐ Launch

Total: 3-4 days

10. COLORS AND TYPOGRAPHY

Color palette:

```
--bg-deep:      #0A0E1A    (deep navy blue)
--bg-mid:       #1A1F2E
--text-primary: #F5F1E8    (warm cream)
--accent-gold:  #D4AF37    (hand light)
--accent-cyan:  #5EEAD4    (butterfly glow)
--accent-pink:  #F0ABFC    (butterfly transform)
```

Typography:

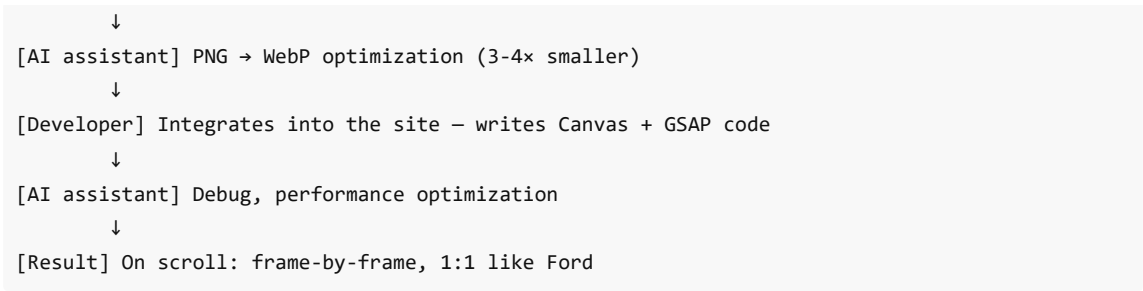
- Hero title:** Fraunces (free Google Font, serif)
- Body:** Inter or Geist Sans (free)
- UI elements:** JetBrains Mono (free)

11. CONFIRMED DECISIONS

Question	Decision
Asset source	AI video generation (Runway / Sora / Kling)
Hosting	Vercel (Free tier)
Animation style	Real human silhouette
Sequence length	240 frames (4-second scroll)
Site language	English only
Asset producer	Developer
Code author	Developer
AI assistant role	Debug, FFmpeg, optimization (does not write code)

12. VIDEO → SCROLL FRAME-BY-FRAME FLOW

```
[Developer] AI video generation (Runway/Sora/Kling)
      ↓
[AI assistant] Extracts frames via FFmpeg script (240 PNGs)
```

Important note:

- A `<video currentTime>` tag is **not used** — it works poorly
- The video is used **as a source only**, then split into frames
- Each frame is stored on the site as a separate image
- This is the standard approach used by major companies like Apple, Ford, BMW

13. INSPIRATION / REFERENCE LINKS

- **Apple AirPods Pro reveal** — the best example
- **Ford Mustang** "Rediscover the Classics" — direct reference
- **Wieden+Kennedy "The Boat"** — long-scroll storytelling
- **Stripe.com Annual Reports** — text + image sequence integration

14. BUDGET

Single expense — AI video tool subscription:

Item	Price	Duration
Runway Standard plan	\$15	1 month
Hosting (Vercel Hobby)	\$0	Free
Fonts (Google Fonts)	\$0	Free
Stock SVG (butterflies — if needed)	\$0	Pixabay/Freepik free

TOTAL: \$15 (one-time, 1-month subscription)

Alternatives (optional):

- ChatGPT Plus + Sora — \$20/month
- Kling AI Pro — \$10/month
- Hailuo — free (with limits)

15. DEADLINE AND TIMELINE

Key dates:

Day	Date	Phase
1	2026-05-14	AI video + Setup + Frame pipeline

2	2026-05-15	Core animation (Canvas + GSAP)
3	2026-05-16	Polish + Responsive + Deploy
4 (buffer)	2026-05-17	Final QA + Launch

Total: 3-4 days — starting today.

Milestones:

#	Milestone	Date
M1	Concept approved	2026-05-14
M2	AI video ready + setup	2026-05-14
M3	Core animation working	2026-05-15
M4	Polish + responsive done	2026-05-16
M5	Production launch	2026-05-17

16. TEAM AND RESPONSIBILITIES

3 roles:

Role	Responsibility
Project Owner	Approve the concept, accept milestones, final quality review
Developer	All technical work: AI video generation, image generation, Next.js, GSAP, Canvas, components, deploy
AI Assistant (Claude)	Only: debug, FFmpeg scripts, video → frame pipeline, WebP optimization, performance tuning

What the AI assistant does NOT do:

- Does not write code (the developer does)
- Does not make design decisions
- Does not deploy automatically

What the AI assistant DOES do:

- Debugs code (identifies and explains errors)
 - Prepares FFmpeg commands (video → frames)
 - Writes WebP optimization scripts
 - Analyzes performance issues and recommends fixes
 - Answers technical questions
-

17. RISKS AND MITIGATIONS

Technical risks:

Risk	Likelihood	Impact	Mitigation
AI video quality is poor	HIGH	High	3-5 iterations, different prompts
Slow load on mobile	Medium	High	Adaptive quality, smaller mobile sequence
iOS Safari Canvas bugs	Low	Medium	Polyfill and testing, ImageBitmap API
240 frames — RAM exhausted	Low	Medium	Progressive loading, frame recycling
FPS drops	Medium	Medium	will-change, offscreen canvas, throttle

Project risks:

Risk	Likelihood	Impact	Mitigation
3-4 day deadline is tight	HIGH	High	Scope is very short (hero only), particle layer optional
AI video not ready on Day 1	Medium	High	Test multiple tools in parallel (Runway + Kling)
Asset quality insufficient	Medium	High	Quality confirmation on Day 1 — if delayed, reduce scope

Risk mitigation strategy:

1. **Minimal scope** — hero only, no extra features
2. **Parallel work** — asset generation runs alongside setup
3. **Fallback** — particle layer/Lenis optional, can be added post-launch
4. **Early quality validation** — if AI video is not confirmed on Day 1, the deadline is shifted

18. SUCCESS METRICS (KPI)

Technical KPI:

Metric	Target	Measurement tool
LCP (Largest Contentful Paint)	< 2.5s	Vercel Speed Insights / Lighthouse
TTI (Time to Interactive)	< 3.5s	Lighthouse
FPS (during scroll)	≥ 60 FPS	Chrome DevTools Performance
Page weight	< 15 MB (desktop) / < 6 MB (mobile)	Lighthouse
Lighthouse Performance Score	> 85	Lighthouse
Cross-browser support	100% (Chrome, Safari, Firefox, Edge)	Manual test

User experience KPI:

Metric	Target	Measurement tool
Scroll completion rate	> 70%	Vercel Analytics scroll depth

Average time on page	> 30 seconds	Analytics
Bounce rate	< 50%	Analytics
CTA click-through (LEARN MORE → .net)	> 15%	Click tracking

Business KPI:

Metric	Target
Social shares	> 500 in the first month
Traffic referred to butterflychallenge.net	> 1000 referrals/month

19. NEXT STEPS

Today (2026-05-14):

- ☐ Approve the concept
- ☐ **Subscribe to Runway Standard** (\$15)
- ☐ **AI video generation** (developer)
- ☐ **Next.js project setup** (developer)

Days 2-4 (2026-05-15 — 2026-05-17):

- ☐ **FFmpeg pipeline** — 240 WebP frames (AI assistant)
- ☐ **ScrollSequence component** (developer)
- ☐ **GSAP ScrollTrigger** integration (developer)
- ☐ **Polish + mobile** (developer)
- ☐ **Vercel deploy** (developer)
- ☐ Launch